Text Mining
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Alex Brandsen

Universiteit Leiden
The Netherlands

Data Archiving and Networked Services
DANS

TDCC-SSH
Social Sciences & Humanities

Topics of this lecture

- What is text mining?
- Why should we use text mining?
- Sequence labelling (Named Entity Recognition)
- Machine Learning and LLMs
- Practical

What is Text Mining?

Deriving information from semi- or unstructured text collections

What is Text Mining?

Examples:

- Search engines
- Spam filters
- Translation
- Turnitin
- Voice assistants
- Customer service bots
- LLMs

Why Use Text Mining?

- Too much text!
- Use computational tools to find or extract useful information
  - **Search engines**: find documents for a research question
  - **Classification**: automatically label papers/reports with subject
  - **Information extraction**: automatically get all c14 dates from corpus
  - **Sentiment analysis**: what do people think of heritage sites?

Why Use Text Mining?

Make our data more FAIR

- Findable: being able to find data relevant for your research
- Accessible: having access to data
- Interoperable: common, machine-readable standards
- Re-usable: use the data for new research!

Text Mining Applications

Types

- **Text classification**
  - input is collection of texts, output is label(s) per text
  - *spam categorisation*
- **Sequence labelling**
  - input is text, output is label for each word
  - *named entity recognition*
- **Sequence-to-sequence learning**
  - input is text, output is a transformation of that text
  - *machine translation*
- **Text Generation**
  - input is text, output is predicted continuation of that text
  - LLMs

Named Entity Recognition
(NER, sequence labelling)

Place (not a period)

… the excavation in Swifterbant was …

… fragments of flint nearby …

… dated to the Iron Age. Other finds …

artefact (not mr. Flint)

Why NER?

Extracting all entities useful for further research:

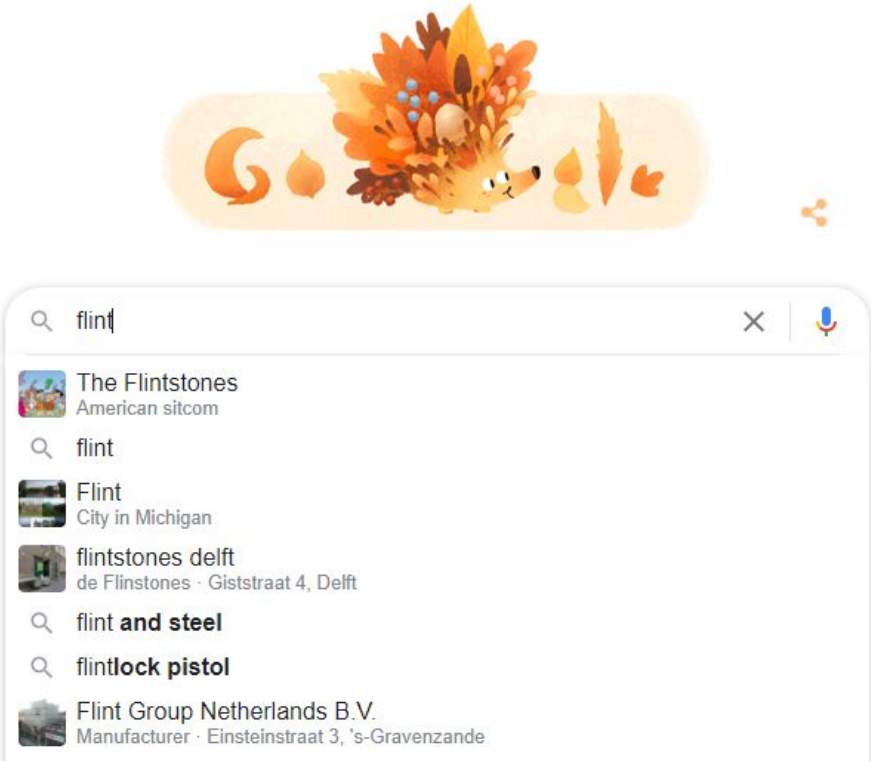
- Overview of time periods mentioned -> temporal range
- Overview of places mentioned -> research area
- Overview of artefacts mentioned -> type of site

- Pattern mining: using all entities of large collection to detect patterns in our data

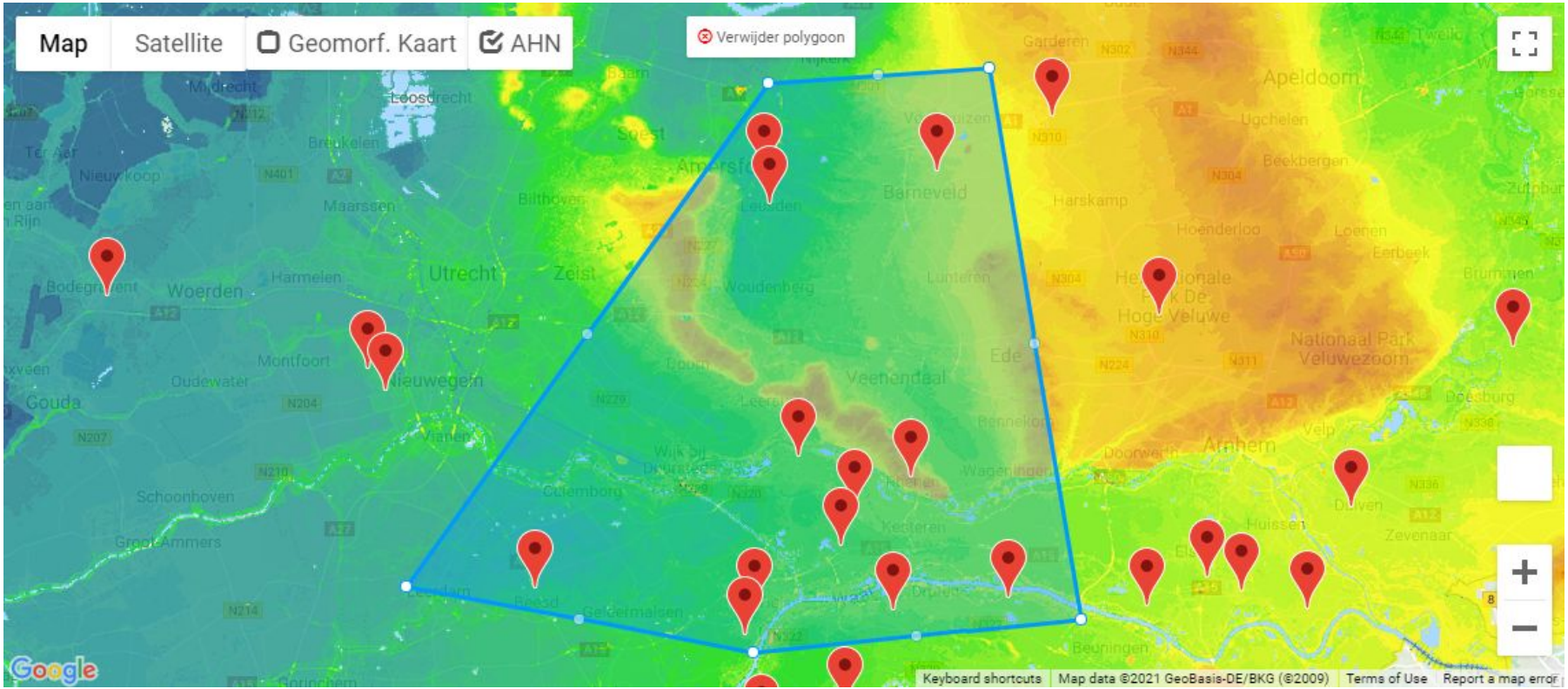
- Search Engines

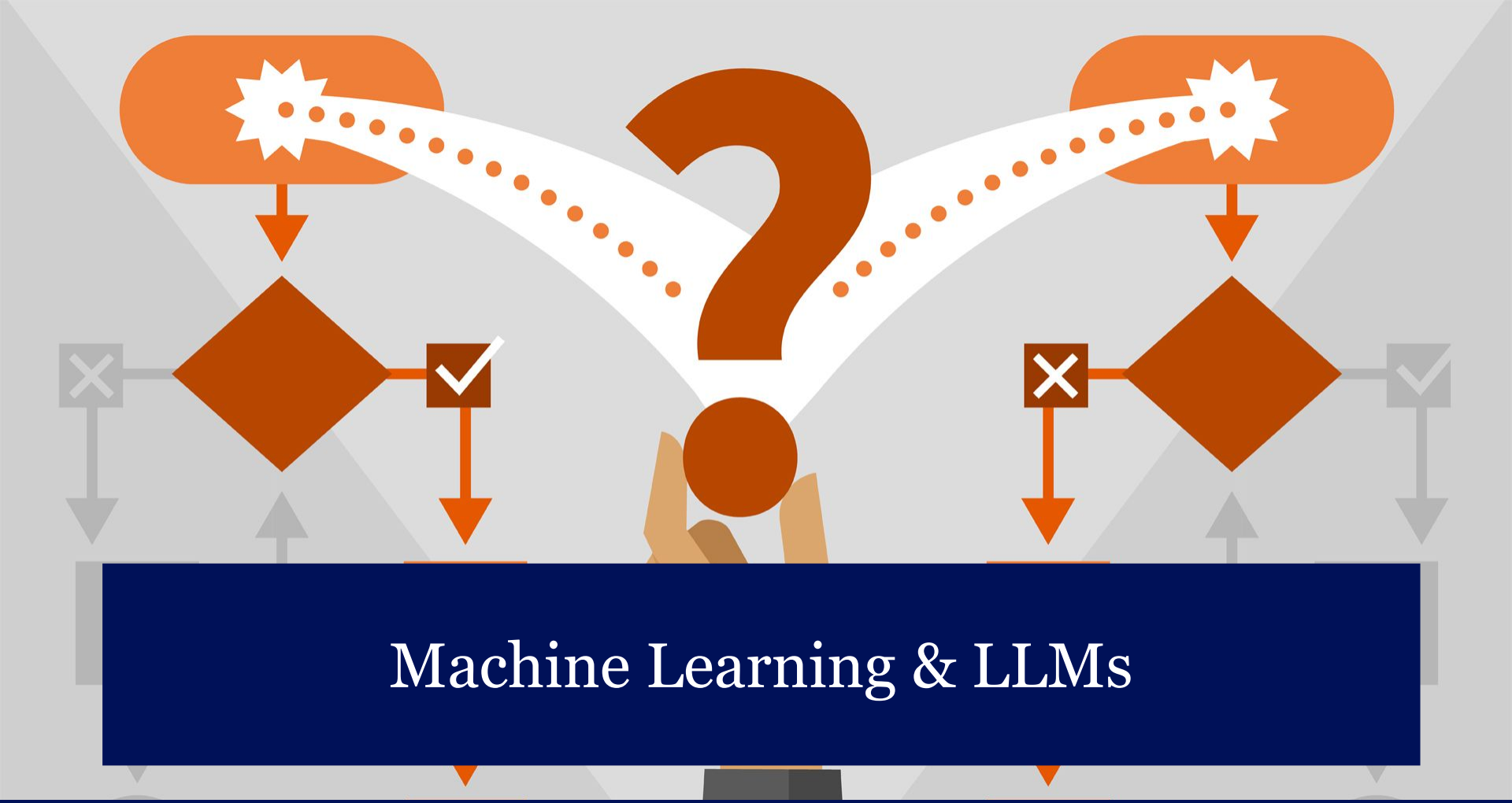
Information Retrieval

Methods to retrieve set of documents relevant to a query



Information Retrieval: AGNES





Machine Learning & LLMs

Transfer Learning with BERT

(Bidirectional Encoder Representations from Transformers)

- Pre-train neural network on large generic dataset
- Model ‘learns’ language (by trying to predict words)
- Fine-tune model on small labeled dataset to do classification
- An ‘early’ LLM, precursor to GPT architecture
- Uses the same Transformer method as GPT

Transfer Learning with BERT

(Bidirectional Encoder Representations from Transformers)

- But: not very good at archaeology…
- Almost no archaeology in original training data
- So we took BERT models, and gave it lots of archaeology data
- Created ArchaeoBERT models
- Performance on archaeo-NER significantly increases

Things are moving fast

- 2017: CRF (traditional machine learning)
- 2018-2019: BERT released, state of the art
- 2020: BERT becomes standard baseline approach
- 2020: laughing at terrible GPT-2 performance:

Things are moving fast

- 2022: GPT-3 released, instant hype, state of the art
- 2023: LLMs outperform on many tasks
- 2024: Image generation still struggles with candy hearts…

- 2025: Perfect candy hearts, corporate LLM hype continues
- 2026: More and more focus on ethical AI, anti-AI sentiment increases

LLM applications in archaeology

- General models are bad at answering archaeology questions
- But can be used to do certain tasks (e.g. coding!)

- Example: extracting knowledge graphs from text
  - Took letters about illegal antiquities trade
  - Fed them into GPT-3, prompt it to extract graph (json)
  - Combine graphs from letters
  - Look into connections via network techniques
  - Find hidden / new actors in the trade network

https://open-research-europe.ec.europa.eu/articles/3-100/v1

LLMs - Ethical Considerations

- Works well, but at what cost?
- Huge biases in data: biases in output
  - Reinforces societal stereotypes and prejudices
- Data unethically sourced: privacy and copyright issues
- Worker exploitation
  - Content filter: labelling descriptions of extreme violence and other horrific situations (similar to FB content moderators)
  - Outsourcing to global South for poverty wages
- Not open source
- Companies use our data to improve commercial product
- Huge environmental impact

LLMs - Current Developments

- Drive for open source / ethical models (Mistral, EuroLLM)
- Reducing size of models (model distillation)
- Explainability (quite difficult)
- Language specific LLMs
  - Dutch: GPT-NL

Questions?

Any questions about any of the subjects?

Practical

Instructions

Programming in Python (no knowledge needed!)

- Go to alexbrandsen.nl/archon
- This opens Google Colab, an interactive way to code
- (you might need to login / create a google account)
- Read the instructions and run the code
- Fill in information/code where needed